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Revised genetic diagrams for natural gases based on a global dataset of >20,000 samples

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ABSTRACT

The origin of natural gases, in particular those containing methane (CH_4 or C_1), ethane (C_2H_6 or C_2), propane (C_3H_8 or C_3) and carbon dioxide (CO_2), is commonly interpreted using binary genetic diagrams of $\delta^{13}C$-C_1 versus $C_1/(C_2+C_3)$, $\delta^{13}C$-C_1 versus δ^2H-C_1 and $\delta^{13}C$-C_1 versus $\delta^{13}C$-CO_2. These diagrams are empirical, but their currently used genetic fields were proposed around 30–40 years ago based on geographically and geologically limited datasets of tens to few hundreds gas samples. As a result, many recently collected gas samples plot outside of accepted genetic fields making these genetic diagrams partly inadequate for the purpose of gas interpretation. Here, we update the genetic diagrams using geochemical and geological data on 20,621 gas samples from a variety of geographical areas (76 countries and territories on six continents) and geological habitats (conventional and unconventional petroleum reservoirs, petroleum seeps and mud volcanoes, gas hydrates, volcanic/geothermal/hydrothermal manifestations, seeps and groundwater in serpentinized ultramafic rocks, aquifers, freshwater and marine sediments, igneous and metamorphic rocks). The revision includes genetic fields for primary microbial gases from CO_2 reduction and methyl-type fermentation, secondary microbial gases generated during petroleum biodegradation, thermogenic and abiotic gases. The genetic field of thermogenic gases now includes early mature ($\delta^{13}C$-C_1 as low as −75‰) and very late mature ($\delta^{13}C$-C_1 around −15‰) gases recently recognized in various petroleum systems. Abiotic C_1 is not necessarily ^{13}C-enriched ($\delta^{13}C > -20$‰) as was often considered in the past. The $\delta^{13}C$ values of abiotic C_1 can be as negative as around −50‰, although a minor component of biotic (microbial or thermogenic) C_1 is often associated with abiotic gas. In addition, the diagrams display molecular and isotopic changes that accompany post-generation processes of mixing, migration, biodegradation, thermochemical sulphate reduction and oxidation. The proposed diagrams cover the vast majority of hydrocarbon-containing gases currently known to exist in nature, are the most comprehensive empirical gas genetic diagrams published to date, and thus represent an essential tool for interpretations of natural gases. Still, holistic integration of geochemical and geological data is necessary to better interpret the origin of natural gases and processes that affected their composition.

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1. Introduction

Earth scientists routinely analyse molecular and isotopic composition of natural gases to interpret their origin (biotic versus abiotic, microbial versus thermogenic), maturity (generation temperature or thermal stress) and alterations. For hydrocarbon-containing gases, such interpretations are based primarily on the genetic diagrams using molecular ratios of alkanes (methane CH_4 or C_1, ethane C_2H_6 or C_2, propane C_3H_8 or C_3), stable carbon ($^{12}C/^{13}C$ or $\delta^{13}C$) and hydrogen ($^1H/^2H$ or δ^2H) isotope composition of C_1 and $\delta^{13}C$ of carbon dioxide (CO_2). These binary diagrams include $\delta^{13}C$-C_1 versus $C_1/(C_2+C_3)$ proposed by Bernard et al. (1977), $\delta^{13}C$-C_1 versus δ^2H-C_1 proposed by Schoell (1983) and Whiticar et al. (1986), and $\delta^{13}C$-C_1 versus $\delta^{13}C$-CO_2 proposed by Gutsalo and Plotnikov (1981). For example, gas dominated by C_1 with $\delta^{13}C$ values <−55‰ is typically considered to have microbial origin, while gas enriched in C_{2+} alkanes and with C_1 that has $\delta^{13}C$ values from −30‰ to −50‰ is typically interpreted as thermogenic (e.g., Milkov, 2005; Strąpoć et al., 2010; Tassi et al., 2012; Pytlak et al., 2016; Schovsbo and Nielsen, 2017; Wu et al., 2017). Abiotic C_1 is commonly assumed to be ^{13}C-enriched, with $\delta^{13}C$ values >−20‰ (e.g., Hunt, 1996).

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(1983) is used mostly for interpretation of petroleum reservoir gases (e.g., Norville and Dawe, 2007; Milkov et al., 2007; Saberi and Rabbani, 2015; Loegering and Milkov, 2017) while the version of Whiticar et al. (1986) is used more widely in environmental studies (e.g., Nicot et al., 2017). Milkov (2011) added the genetic field of significant secondary microbial C_1. Etiope and Sherwood Lollar (2013), Etiope and Schoell (2014) and Etiope (2017a) added new wide field of abiotic C_1 to this genetic diagram.

Gutsalo and Plotnikov (1981) published the first gas genetic diagram based on $\delta^{13}C$ of C_1 and CO_2. They distinguished fields of endogenic (abiotic), biogenic (microbial) and thermogenic gases. Whiticar (1999) distinguished bacterial (primary microbial) gases generated from CO_2 reduction and from methyl-type fermentation. Milkov (2011) added secondary microbial gases to that genetic diagram.

3. Methods

3.1. Global database

We compiled a global database of 20,621 gas samples using >600 published papers and reports as well as publicly available government databases. The database includes 7428 fossil fuel samples (after quality control of many samples based on original references) from the global inventory of gas geochemical data of Sherwood et al. (2017), which we used as a starting point for our database. Not all samples have complete geographical, geological and geochemical information available for them. Thus, statistical parameters below are based only on samples for which the relevant information is available and recorded.

The database includes samples from 76 countries and territories on six continents (Fig. 2), from both onshore (including lakes) and offshore environments. Most gas samples come from the USA (5717 or 28%), China (5548 or 27%), Russia (1470 or 7%), Canada (1106 or 5%) and Australia (767 or 4%). The database is biased towards natural settings highly enriched in hydrocarbon-containing gases. There are 13,761 samples (67% of all samples in the database) from conventional oil and gas reservoirs and 3932 (19%) samples from unconventional petroleum reservoirs (coal bed methane or CBM, tight sandstones, shale and gas hydrate reservoirs). The database also includes samples from petroleum seeps and mud volcanoes, volcanic/geothermal/hydrothermal fluid manifestations, seeps and groundwater in serpentinized ultramafic rocks (peridotites), aquifers, freshwater sediments (lakes, swamps), marine sediments, and non-sedimentary hard rocks. The pie chart in Fig. 3 summarizes the types of gas habitats in the database. The subsurface depth of samples with available information varies from 0 m (surface or seafloor) to 8600 m (Fig. 4). Gases from conventional petroleum reservoirs come from wellheads and formation evaluation tools such as drill stem testers (DST) and modular dynamics testers (MDT). We deliberately excluded gases collected during drilling (mudgas logging, headspace and IsoTube samples) to avoid issues with fractionation, air dilution, fluid circulation and drillbit metamorphism (Wenger et al., 2009). For CBM and shale gases, both gases collected at wellheads during production and gases collected during samples desorption were used. We recognize that the composition of produced gases may change over time, especially in CBM wells (Mastalerz et al., 2017; Niemann and Whiticar, 2017), and we included gases from different periods of production, when possible.

Although the database is a large body of geochemical and geological information (>690,000 entries), in this study we focus on molecular composition of main hydrocarbon gases (expressed as the ratio $C_1/(C_2 + C_3)$), carbon and hydrogen isotopic composition ($\delta^{13}C$ and $\delta^{2}H$) of C_1 and carbon isotopic composition ($\delta^{13}C$) of CO_2. The frequency distributions of these gas properties in our database are shown as histograms in Fig. 5, and Table 1 lists the main statistics on geochemical composition of gases from various geological habitats. Isotopic composition of carbon is presented in parts per thousand (per mil, ‰) values referenced to the Pee Dee Belemnite (PDB or Vienna-PDB) standard. Isotopic composition of hydrogen is presented in ‰ values referenced to the Standard Mean Ocean Water (SMOW) standard. As the range of properties among the samples is very large relative to precision and accuracy of measurements (for example, about 155‰ for $\delta^{13}C$-C_1 for 17,683 gas samples (Fig. 5B) while most labs report precision and accuracy of measurements around ±0.3–0.5‰), we assume that the quality and inter-laboratory reproducibility of gas measurements are not a significant issue for the purposes of this study. Still, we corrected the datasets which were obviously internally inconsistent, as, for example, the old data from the Soviet Union (Milkov, 2010).

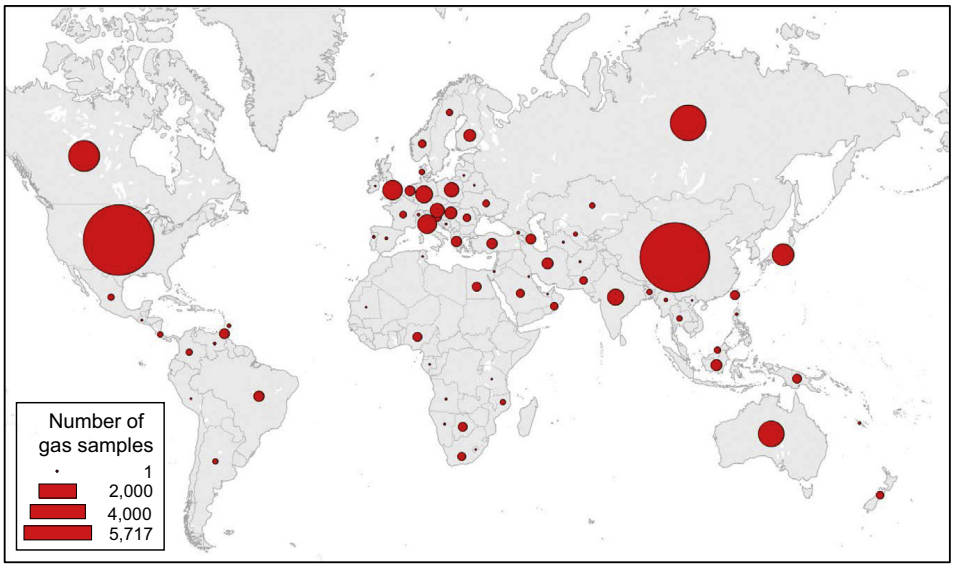

Fig. 2. Map showing the countries and territories with gas samples included in the studied global dataset. The size of circles is proportional to the number of gas samples.

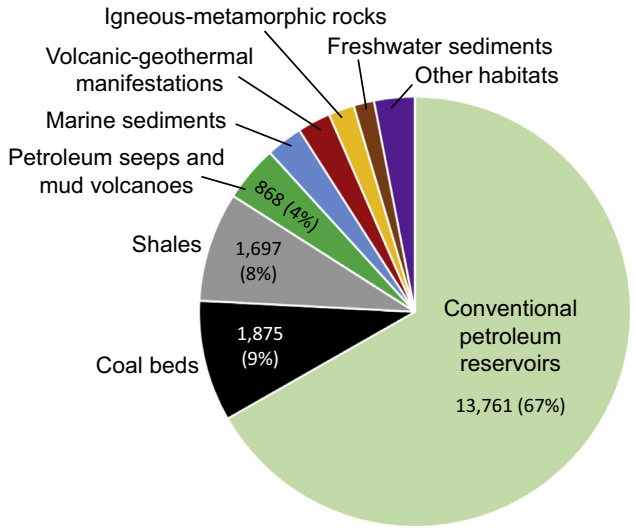

Fig. 3. Pie chart of the habitats of natural gases in the global database (20,621 gas samples) used in this study. "Other habitats" include samples from gas hydrates, seeps and groundwater in serpentinized ultramafic rocks, tight reservoirs, aquifers and salt mines.

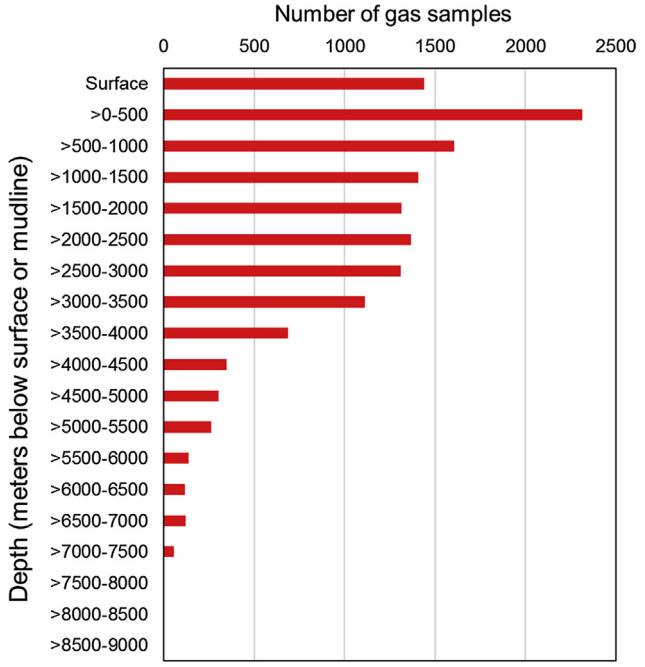

Fig. 4. Subsurface depth of gas samples in the database. The depth information is available and displayed for 13,890 samples (67% of the samples in our database). There are only four samples from >7500 to 9000 m, so they are not visible in the plot.

3.2. Data interpretation

The genetic diagrams of Bernard et al. (1977), Gutsalo and Plotnikov (1981), Schoell (1983) and Whiticar et al. (1986) are empirical. We revised and added genetic fields in these diagrams based on our interpretation of origin of gases in our large global database. Although the interpretation is subjective, it is internally consistent and robust as we integrated geochemical data on gases with geological habitats of gases, including the lithology of host rocks, regional geology, petroleum systems, temperature etc. We assigned "dominant" gas origin (e.g., microbial, thermogenic, abiotic) to 8108 gas samples. This is ~39% of samples in our database. Although many gases in the database have insufficient geochemical and/or geological information to conclusively infer their origin, the majority of natural gases are mixtures of gases of different origin or gases affected by post-generation processes. Below we list the main geochemical and geological criteria for defining the origin of natural gases and post-generation processes that affected them.

Primary microbial gases were recognized based mainly on the geochemical characteristic of having only C_1, C_2, and C_3 as hydrocarbon gases (C_2 and C_3 are present in much smaller amounts than C_1) and having C_1 enriched in ^{12}C. This is consistent with both laboratory experiments and observations in natural environments that microbes can generate only these three hydrocarbon compounds (Oremland et al., 1988; Hinrichs et al., 2006). Geological setting of the gases provided additional important criteria to recognize primary microbial gases (e.g., lack of oil in the reservoirs or sediments) and to distinguish gases from predominantly CO_2-reduction (recent marine sediments, reservoir gases) and from predominantly methyl-type fermentation (recent freshwater sediments) (Whiticar, 1999).

Thermogenic gases were recognized based on the presence of all methane homologues from C_1 to C_5 (pentanes) and the semi-linear plot of $\delta^{13}C_n$ values versus 1/n (Chung et al., 1988). Such pure thermogenic gases were distinguished from gases that experienced biodegradation, modification during thermochemical sulphate reduction (TSR) and mixing with gases of other origin (such as primary microbial gases).

Secondary microbial gas (mostly C_1) is produced by microbes during petroleum biodegradation. This gas is most often mixed with oil-associated biodegraded thermogenic gas, which originally was part of the biodegradation feedstock. Although it is difficult to estimate if secondary microbial gas dominates in the gas mixture, we assigned the secondary microbial origin to gases that have $\delta^{13}C$-CO_2 exceeding +2‰ (Pallasser, 2000; Milkov, 2011, 2018) and associate with clearly biodegraded oil and gas (e.g., removed *n*-alkanes in oils (Head et al., 2003), C_3 enriched in ^{13}C, increased *i*-C_4/*n*-C_4 ratio (James and Burns, 1984)).

The abiotic origin of the hydrocarbon-containing gases produced by post-magmatic and Fischer-Tropsch type reactions, i.e. CO_2 hydrogenation or Sabatier reaction (Etiope and Sherwood Lollar, 2013) was recognized on the basis of multiple lines of evidence. These include the geological settings and habitats that do not have mature organic-rich source rocks as well as various geochemical indicators. Abiotic gases typically occur in Precambrian crystalline shields, igneous batoliths, geothermal fluids, hyperalkaline springs and aquifers in serpentinized ultramafic rocks in ophiolites or peridotite massifs (Botz et al., 1996; Potter and Konnerup-Madsen, 2003; Sherwood Lollar et al., 2006; Tassi et al. 2012; Etiope and Schoell, 2014; Etiope et al., 2017 and references therein). Geochemical evidence for abiotic origin include (a) a specific combination of $\delta^{13}C$ and $\delta^{2}H$ values of C_1 (outside and only partially overlapping the thermogenic field; Etiope, 2017a; Etiope et al. 2017); (b) the inverse isotopic trend of normal alkanes (i.e., decreasing ^{13}C content in alkanes with longer chains, $\delta^{13}C$-$C_1 > \delta^{13}C$-$C_2 > \delta^{13}C$-C_3, which is typical, although not exclusive, of abiotic polymerization of alkanes; Sherwood Lollar et al., 2008); (c) the Schulz-Flory distribution of the alkanes controlled by chain growth probability factor for abiotic stepwise polymerization where ($C_n + 1/C_n$) is approximately constant (C_n is the concentration in mole units); and (d) considerable amounts of hydrogen gas (H_2), typically produced by serpentinization of olivine-rich rocks and radiolysis (Etiope and Sherwood Lollar, 2013). While CO_2 is the dominant abiotic gas in volcanic geothermal systems, C_1 is generally the main gas (>50 vol%, often together with H_2) in ultramafic rocks and crystalline shields (Etiope and Schoell,

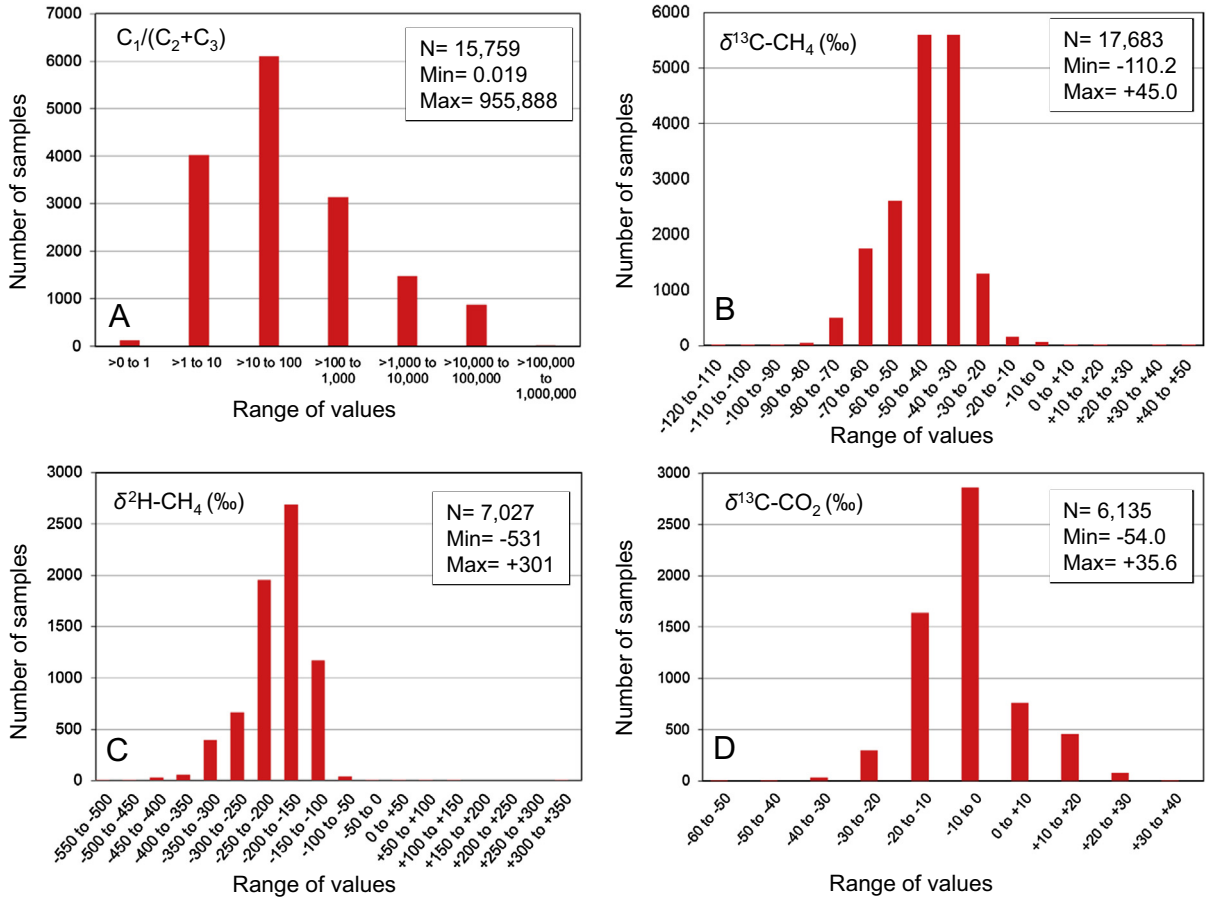

Fig. 5. Distribution and key statistical parameters for (A) $C_1/(C_2 + C_3)$, (B) $\delta^{13}C$-C_1, (C) δ^2H-C_1 and (D) $\delta^{13}C$-CO_2 values for gas samples in the database. Note that not all samples have all four geochemical parameters measured.

Table 1
Number and range of values of subsurface depth and geochemical composition of gas samples from various geological habitats in our dataset. The "number of samples" reflects all samples from the particulate habitat in the dataset, even though not all samples have all geochemical data available for them. Notes: mbs = meters below surface, mbml = meters below mudline, all δ values are in per mil (‰) and rounded to the nearest whole number.

Habitat	Number of samples	Depth (mbs or mbml)		$C_1/(C_2 + C_3)$		$\delta^{13}C$-CH_4		δ^2H-CH_4		$\delta^{13}C$-CO_2	
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
Conventional petroleum reservoirs	13,761	11	8600	0.02	100,000	−91	−3	−477	−49	−46	+33
Coal beds	1875	18	4424	0.75	240,600	−88	−17	−415	−75	−54	+26
Shales	1697	2	4480	0.18	100,000	−82	−21	−336	−123	−27	+27
Petroleum seeps and mud volcanoes	868	0	<100	2	263,071	−82	−16	−348	+124	−34	+36
Volcanic / geothermal / hydrothermal fluids	526	0	3235	2	100,000	−72	+45	−377	+301	−29	+6
Igneous and metamorphic rocks	399	0	5684	0.08	100,000	−83	+4	−470	−50	−34	−2
Marine sediments	565	0	1998	33	179,770	−110	−33	−369	−101	−50	+15
Freshwater sediments	316	5	22	9	955,888	−86	−31	−531	−105	−26	+2
Gas hydrates	243	0	920	2	136,620	−75	−40	−242	−115	−30	+26
Seeps/aquifers in serpentinized ultramafics	175	0	<200	1	100,000	−70	+15	−428	−68	−32	−7
Tight petroleum reservoirs	117	647	4321	1	264	−45	−26	−207	−133	−32	−3
Aquifers and groundwater	70	12	1520	17	100,000	−75	−20	−403	−141	−29	−12

2014). As mentioned earlier, abiotic gas, especially in geothermal systems and serpentinized ultramafic rocks, is often mixed with variable, generally minor, amounts of thermogenic and/or microbial gas. The "dominant" abiotic character is however assessed on the basis of the diagnostic tools described above and following interpretations in published works (e.g. Etiope et al., 2017 and references therein).

Fig. 6 summarizes distributions of $C_1/(C_2 + C_3)$, $\delta^{13}C$-C_1, δ^2H-C_1 and $\delta^{13}C$-CO_2 for gases with interpreted "dominant" origin. In addition to identifying the origin of gases, we interpreted the processes that affected these gases. Biodegradation was interpreted based on the presence of biodegraded oils, geological setting (e.g., relatively low temperatures and shallow reservoirs) and the ^{13}C-enrichment of relatively more degradable C_3, *n*-butane (*n*-C_4) and *n*-pentane (*n*-C_5) (James and Burns, 1984; Milkov, 2011). TSR was inferred from geological habitat (petroleum fluids in carbonate reservoirs at relatively high temperatures >100 °C) and geochemical evidence (elevated H_2S) (e.g., Worden et al., 1996; Worden and Smalley,

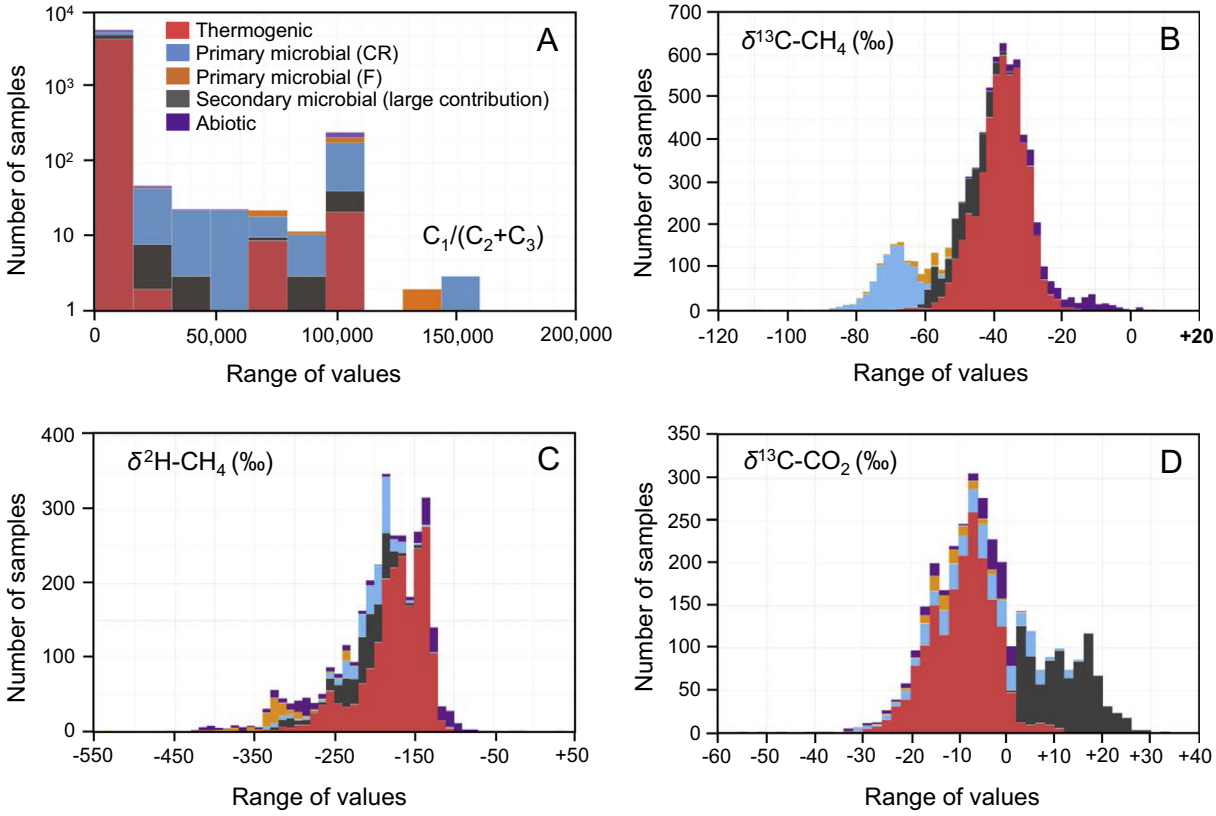

Fig. 6. Distribution of $C_1/(C_2 + C_3)$ (A, bin size 15,927; 6588 samples), $\delta^{13}C\text{-}C_1$ (B, bin size 2‰; 7426 samples), $\delta^{2}H\text{-}C_1$ (C, bin size 10‰; 3244 samples) and $\delta^{13}C\text{-}CO_2$ (D, bin size 2‰; 3304 samples) for samples interpreted as "dominantly" thermogenic, primary microbial (CR - CO_2 reduction, F - methyl-type fermentation), secondary microbial and abiotic gases.

2004). Oxidation was interpreted based on extreme enrichment of carbon and hydrogen in heavy isotopes (e.g., Etiope et al., 2011), considering that microbial oxidation of methane produces variations of $\delta^{13}C$ and $\delta^{2}H$ of C_1 correlated with $\Delta H/\Delta C$ values ~8–9 (Coleman et al., 1981; Kinnaman et al., 2007).

4. Results

The original genetic gas fields proposed by Bernard et al. (1977), Schoell (1980, 1983), Gutsalo and Plotnikov (1981), Whiticar et al. (1986) and Whiticar (1999) were based on relatively small gas datasets (tens to first hundreds samples) restricted to certain geographic areas and subsets of gas habitats and origins. Fig. 7A, Fig. 8A and Fig. 9A show that a large number of samples in our global database plot outside of the originally outlined genetic gas fields. We used our global database to update the diagrams based on $\delta^{13}C\text{-}C_1$ versus $C_1/(C_2 + C_3)$ (Fig. 7), $\delta^{13}C\text{-}C_1$ versus $\delta^{2}H\text{-}C_1$ (Fig. 8), and $\delta^{13}C\text{-}C_1$ versus $\delta^{13}C\text{-}CO_2$ (Fig. 9). First, we reviewed all available geochemical and geological information on all gas samples and determined the likely origin of "dominant" gases as described in the Methods section above (Fig. 7B, Fig. 8B, Fig. 9B). We then outlined revised genetic gas fields on all diagrams (Fig. 7C, Fig. 8C, Fig. 9C) and defined their limits in Table 2. An alternative version of the $\delta^{13}C\text{-}C_1$ versus $\delta^{2}H\text{-}C_1$ diagram, with $\delta^{13}C$ on the x axis, as frequently used in works dealing with abiotic gas (e.g., Schoell et al., 1988; Sherwood Lollar et al. 2008; Etiope and Sherwood Lollar, 2013; Etiope et al. 2017) is provided in Supplementary material. Fig. 7D, Fig. 8D and Fig. 9D demonstrate various processes that affect molecular $(C_1/(C_2 + C_3))$ and isotopic ($\delta^{13}C\text{-}C_1$, $\delta^{2}H\text{-}C_1$, and $\delta^{13}C\text{-}CO_2$) composition of natural gases.

5. Discussion

Our revised diagrams are the most comprehensive empirical gas genetic diagrams published so far. They cover most samples of natural gases published to date, have significantly updated fields of primary microbial and thermogenic gases relative to the original plots (Bernard et al., 1977; Gutsalo and Plotnikov, 1981; Whiticar et al., 1986; Whiticar, 1999) and include genetic fields of secondary microbial gases and abiotic gases. The empirical genetic fields should be treated as general interpretative guidelines. Interpretations of specific gas samples should always be based on the integration of comprehensive geological and geochemical data on these gases.

5.1. Origins of gases

Extension of the genetic field of thermogenic gas is one of the most significant modifications in our updated diagrams. Most previous versions of the diagrams implied that thermogenic gases have C_1 with $\delta^{13}C > -50$‰. Indeed, as source rocks become more thermally mature, they expel C_1 relatively enriched in ^{13}C. However, we noted that very early mature thermogenic gases have $\delta^{13}C\text{-}C_1$ from −55‰ to −73‰ and are more enriched in C_1 than later gases generated in the middle oil window (e.g., gases reported by Rowe and Muehlenbachs (1999) and Schovsbo and Nielsen (2017)). Such early mature thermogenic gases are increasingly recognized in shales based on (a) the presence of C_{4+} gases, which, to our knowledge, cannot be produced by microbes; and (b) $\delta^{13}C$ of C_1-C_5 which plot as a semi-straight line on Chung plot (Chung et al., 1988) suggesting lack of mixing of gases with different origin (Fig. 10). We included early mature thermogenic gases in genetic

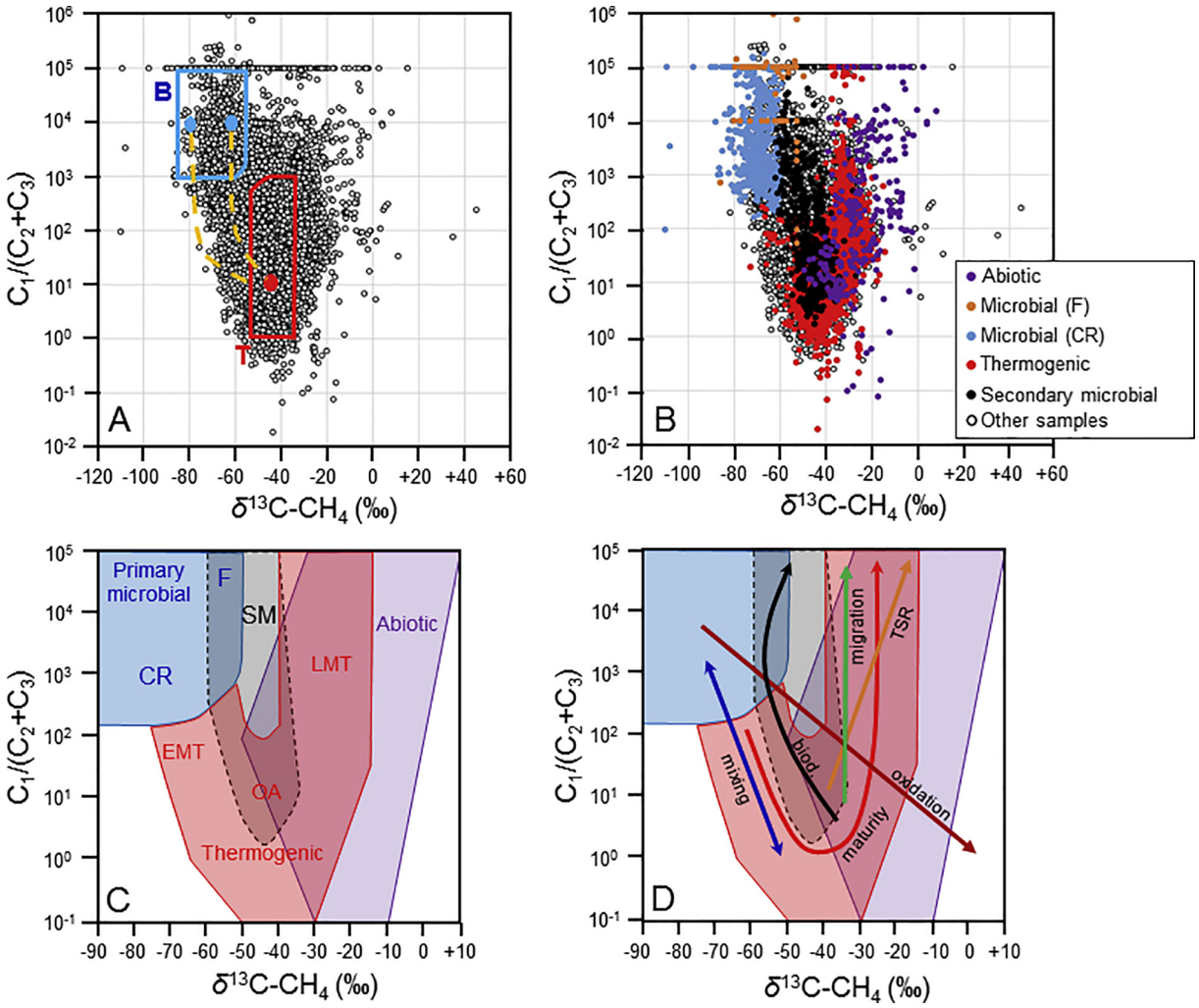

Fig. 7. Genetic diagram of $\delta^{13}C$-C_1 versus $C_1/(C_2 + C_3)$. (A) Data for 13,491 samples from various habitats. The original gas genetic fields and mixing lines of Bernard et al. (1977) are shown with the authors' notations (B – biogenic or microbial, T – thermogenic or petrogenic). Note that many samples plot outside of the originally defined genetic fields. (B) Gas samples shown with the interpretation of their origin based on the integrated geological and geochemical evaluation of the samples and their habitats. (C) Revised genetic fields (CR – CO_2 reduction, F – methyl-type fermentation, SM – secondary microbial, EMT – early mature thermogenic gas, OA – oil-associated thermogenic gas, LMT – late mature thermogenic gas). (D) Processes that affect molecular and isotopic composition of the gases (biod. – biodegradation, TSR – thermochemical sulphate reduction). Mixing of gases with different origin is commonly observed in nature, but, for simplicity, we show only mixing of primary microbial and thermogenic gases in this diagram.

diagrams in Fig. 7C, Fig. 8C and Fig. 9C (and Supplementary material). These early mature thermogenic gases may be misinterpreted as pure primary microbial gases (see, for example, Strąpoć et al., 2010). However, very early mature thermogenic gases may be mixed with primary microbial gases within the source rocks as gas generation likely does not switch from purely microbial process to purely thermogenic process but is a transition. Generation of early mature thermogenic gases is less quantitatively significant than generation of primary microbial gases and especially oil-associated/late mature thermogenic gases. Furthermore, early mature gases are often mixed with later generated more mature thermogenic gases. As a result, early mature thermogenic gases are rarely identified in petroleum reservoirs (although there are examples from both conventional reservoirs (Milkov and Dzou, 2007; Goncharov et al., 2017) and unconventional reservoirs (Lavoie et al., 2016)) and are usually not captured in laboratory pyrolysis experiments.

In contrast to most recently published versions of the Bernard plot (e.g., Pytlak et al., 2016; Wu et al., 2017), our revision of the genetic diagram based on $\delta^{13}C$-C_1 versus $C_1/(C_2 + C_3)$ does not include the areas of gases generated from Type II and Type III kerogen as originally proposed by Faber et al. (1988) based on a limited gas dataset from Germany. The reason for that is that we did not observe a separation of these gases on the plot of $\delta^{13}C$-C_1 versus $C_1/(C_2 + C_3)$ in our global dataset.

The genetic fields of primary microbial gas are largely consistent with the original genetic diagrams. There are reports of primary microbial gas with low C_1/C_2 (as low as ~40) and C_1 highly enriched in ^{13}C ($\delta^{13}C$ as high as –32.6‰) in endoevaporitic settings (Tazaz et al., 2013). However, we did not extend the genetic fields of primary microbial gases to such extreme values because they are rather unique in our global dataset. Future finds of primary microbial gas enriched in C_{2+} gases and ^{13}C may justify the additional revision of this genetic field.

The $\delta^{13}C$-C_1 versus $C_1/(C_2 + C_3)$ diagram on its own is not suitable for identifying abiotic gas, which may have a wide range of $C_1/(C_2 + C_3)$ values. These values depend on the specific abiotic generation mechanism and may involve different degrees of methane polymerization resulting in variable C_{2+} abundance (Etiope and Sherwood Lollar, 2013). The ratio $C_1/(C_2 + C_3)$ is also affected by the type of sampled fluid because differential solubility and migration fractionations may alter the original relative abundance of

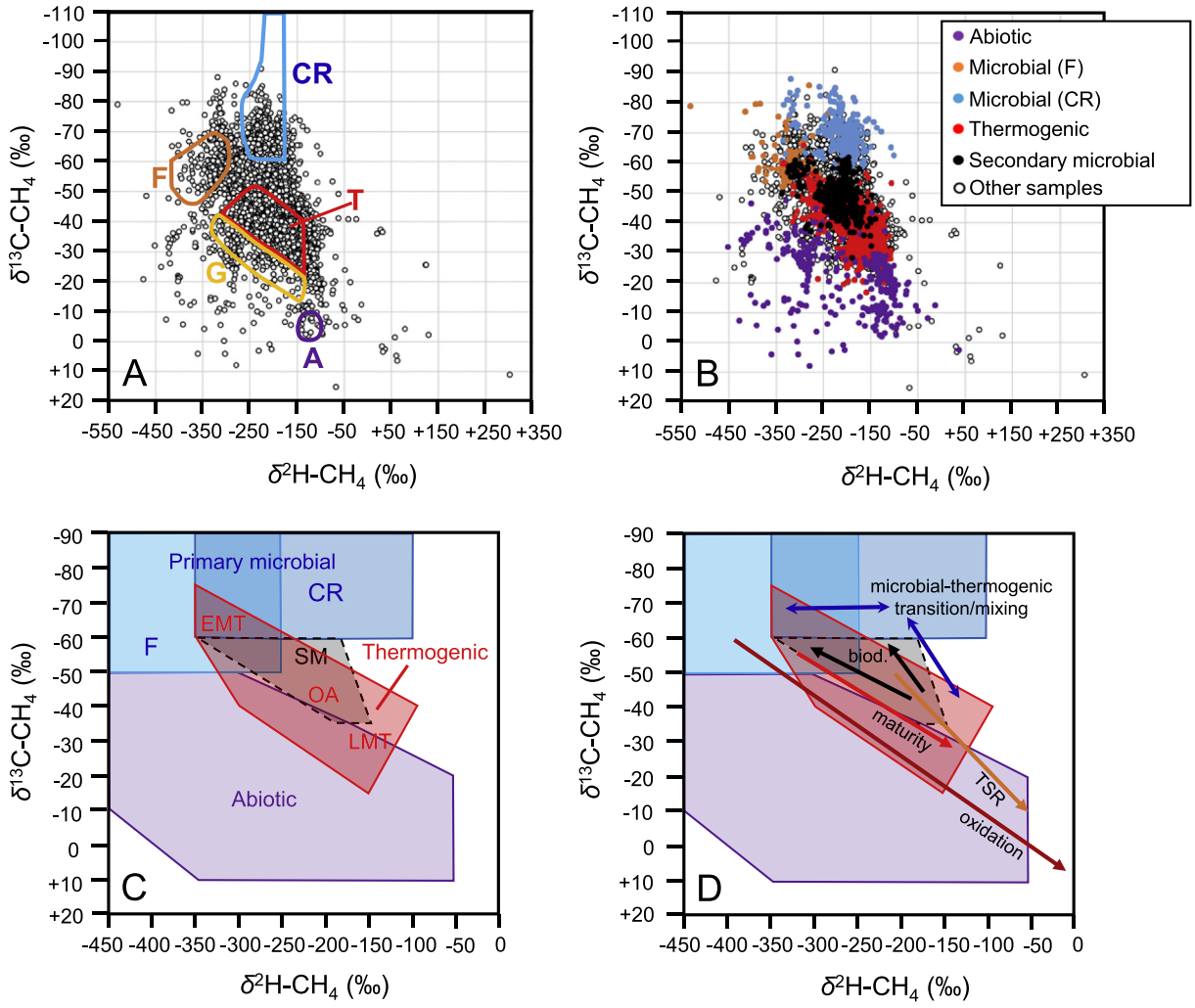

Fig. 8. Methane genetic diagrams based on δ^{13}C-C_1 versus δ^2H-C_1. (A) Data for 6950 samples from various habitats. The gas genetic fields of Whiticar et al. (1986) and Whiticar (1999) are shown with the original notations (CR – CO_2 reduction, F – fermentation, T – thermal or thermogenic, G – geothermal, hydrothermal, crystalline, A – abiogenic). Many samples plot outside of the originally defined genetic fields. (B) Gas samples shown with the interpretation of their origin based on the integrated geological and geochemical evaluation of the samples and their habitats. (C) Revised genetic fields (acronyms as in Fig. 7C). (D) Processes that affect isotopic composition of C_1 in natural gases (acronyms as in Fig. 7D).

alkanes. The genetic field for abiotic C_1 on the diagram of δ^{13}C-C_1 versus δ^2H-C_1 is better defined compared to early genetic fields published before 2013 but is similar to recent revisions by Etiope and Schoell (2014) and Etiope (2017a,2017b). The lower and upper δ^{13}C-C_1 limits of abiotic gas, however, cannot be readily identified. Positive δ^{13}C-C_1 values reported for several abiotic gases may actually result from C_1 oxidation, as discussed in Section 5.2, and more ^{13}C-depleted samples may host biotic (microbial and/or thermogenic) components. As extreme δ^{13}C-C_1 values in the diagram, we used +10‰ (assuming non-oxidation in some ^{13}C-enriched samples from Oman and Greece, Vacquand et al., 2018; Daskalopoulou et al., 2018) and −50‰ (close to the abiotic gas of Driefontein, South Africa Precambrian Shield, considered to have minimal microbial contamination based on multiple geochemical and microbiological analyses, Sherwood Lollar et al., 2006). It is important to note that abiotic C_1 with δ^{13}C as low as −145‰ was obtained in laboratory experiments of Sabatier reaction (Etiope and Ionescu, 2015), which suggests the potential of a wider isotopic range for abiotic C_1 in nature. More generally, the abiotic genetic field should be interpreted as representing dominantly abiotic gas, since, as stated above, many samples seem to have variable, although minor, biotic components. Compared to early views, however, abiotic C_1 is not necessarily ^{13}C-enriched, and it may have δ^{13}C values resembling thermogenic gas. Still, abiotic C_1 has a specific combination of δ^{13}C and δ^2H values that only partially overlaps the thermogenic field (Fig. 8C).

There is significant overlap in δ^{13}C-CO_2 values for gases of primary microbial, thermogenic and abiotic origin on the genetic diagram based on δ^{13}C-C_1 and δ^{13}C-CO_2. However, this genetic diagram is very useful for distinguishing secondary microbial gases, which usually have CO_2 with δ^{13}C > +2‰ (up to +36‰, Tassi et al., 2012) and C_1 with δ^{13}C around −50‰. Some primary microbial gases have δ^{13}C-CO_2 up to +15‰ (Toki et al., 2012) but they can be distinguished from secondary microbial gases based on δ^{13}C-C_1 more negative than −60‰ (Fig. 9C). Some highly mature thermogenic gases have δ^{13}C-CO_2 up to +11‰ (Liu et al., 2018) but they have C_1 significantly enriched in ^{13}C (δ^{13}C of C_1 around −30‰, Fig. 9C). Importantly, the origin of CO_2 associated with dominantly abiotic gas (Table 2) may not be necessarily abiotic, as it may derive (as C_1 precursor via Fischer-Tropsch type, or Sabatier, reactions) from a variety of biotic sources (e.g., Etiope et al., 2017). An interesting aspect is, then, the possible erroneous "inorganic" attribution of CO_2 origin when its δ^{13}C value is around zero (from −8 to +3‰). Volcanologists typically consider this range

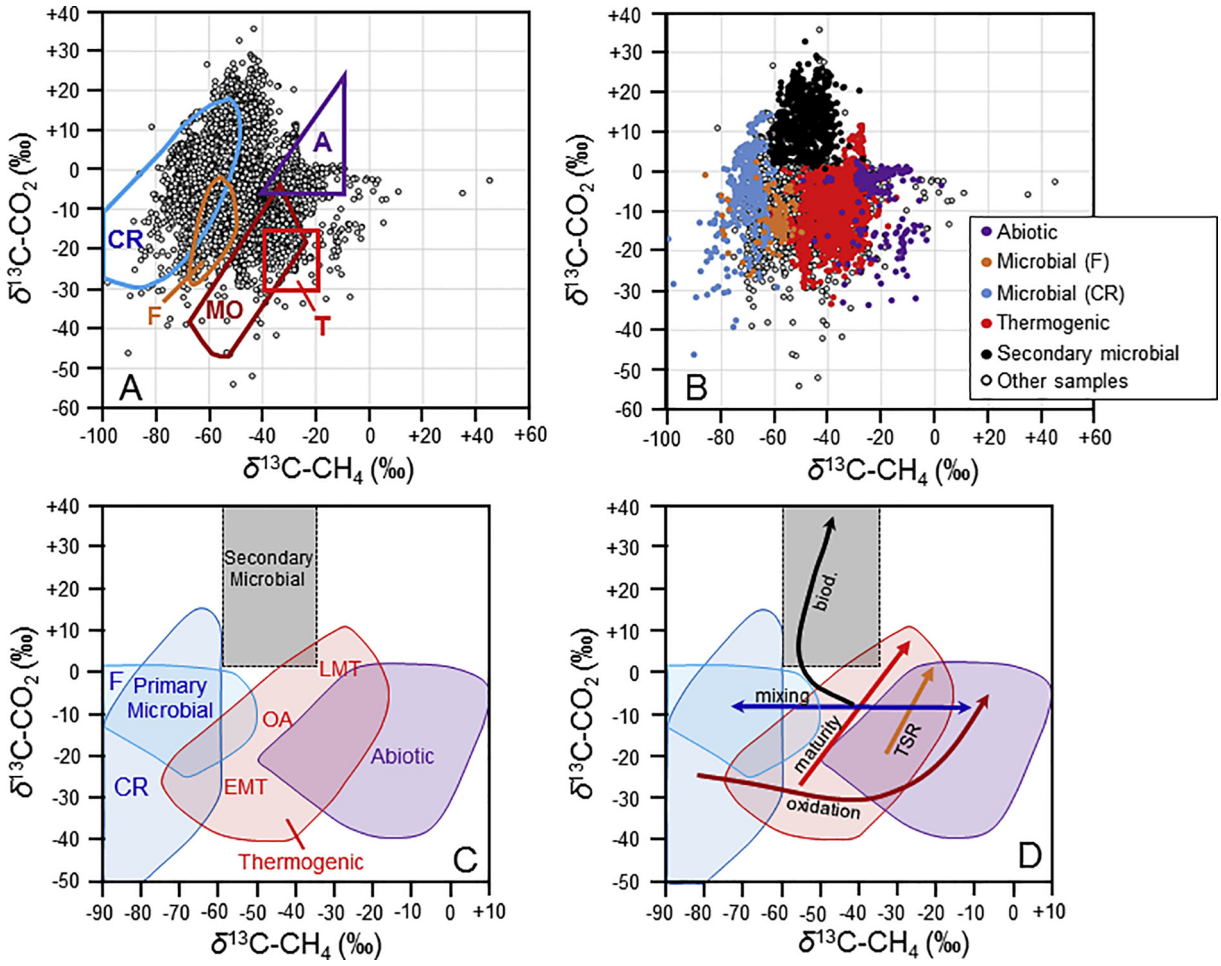

Fig. 9. Genetic diagrams of δ^{13}C-C_1 versus δ^{13}C-CO_2. (A) Data for 5848 samples from various habitats. The original genetic fields of Gutsalo and Plotnikov (1981) (T – thermogenic, A – abiogenic) and of Whiticar (1999) (CR – CO_2 reduction, F – fermentation, MO – methane oxidation) are shown. As in previous diagrams, many samples plot outside of the originally defined genetic fields. (B) Gas samples shown with the interpretation of their origin based on the integrated geological and geochemical evaluation of the samples and their habitats. (C) Revised genetic fields (acronyms as in Fig. 7C). (D) Processes that affect carbon isotopic composition of C_1 and CO_2 in natural gases (acronyms as in Fig. 7D).

Table 2
Minimum and maximum values of geochemical parameters used in the gas genetic fields revised in this study. The extreme values represent genetic boundaries assessed as described in the text, not including fractionation or oxidation, as much as it was possible to evaluate from published data. The genetic fields describe "dominant" components, as most gases (and especially gases with secondary microbial and abiotic origin) often have minor amounts of gas of different origin. Note that for the abiotic gases, CO_2 is not necessarily abiotic as discussed in the text.

Gas genetic fields drawn in diagrams	$C_1/(C_2+C_3)$		δ^{13}C-CH_4 (‰)		δ^{2}H-CH_4 (‰)		δ^{13}C-CO_2 (‰)	
	Min	Max	Min	Max	Min	Max	Min	Max
Primary microbial from CO_2 reduction	200	100,000	−90	−60	−350	−125	−50	+15
Primary microbial from methyl−type fermentation	1000	100,000	−90	−50	−450	−250	−25	+2
Thermogenic	0.1	100,000	−75	−15	−350	−100	−40	+10
Secondary microbial	2	100,000	−60	−35	−350	−150	+2	+40
Abiotic	0.1	100,000	−50	+10	−450	−50	−40	+2

of values as representative of mantle or thermometamorphism. While this is true in mere volcanic-geothermal systems with CO_2 concentrations exceeding ~50 vol%, in C_1-rich sedimentary systems or hybrid sedimentary hosted geothermal (CO_2-C_1) systems, ^{13}C-enriched CO_2 can be due to secondary methanogenesis following hydrocarbon biodegradation (Fig. 9C).

5.2. Processes

Fig. 7D, Fig. 8D and Fig. 9D demonstrate the main processes affecting natural gases. Source rocks initially generate early mature thermogenic gases depleted in C_{2+}. As maturity of source rocks increases, the generated gases become enriched in C_{2+} (oil-associated gases) and then become again depleted in C_{2+} (late mature gases). Maturation of source rocks leads to enrichment of C_1 in heavy isotopes ^{13}C and ^{2}H, and enrichment of CO_2 in ^{13}C.

Gases become more enriched in C_1 (have higher $C_1/(C_2+C_3)$ ratio) during biodegradation because microbes preferentially consume C_{2+} gases (James and Burns, 1984) while secondary microbial C_1 is the terminal product of biodegradation (Zeikus, 1977; Head et al., 2003; Larter et al., 2005) and is added to the gas mixture. Methane can become less enriched or more enriched in ^{13}C relative

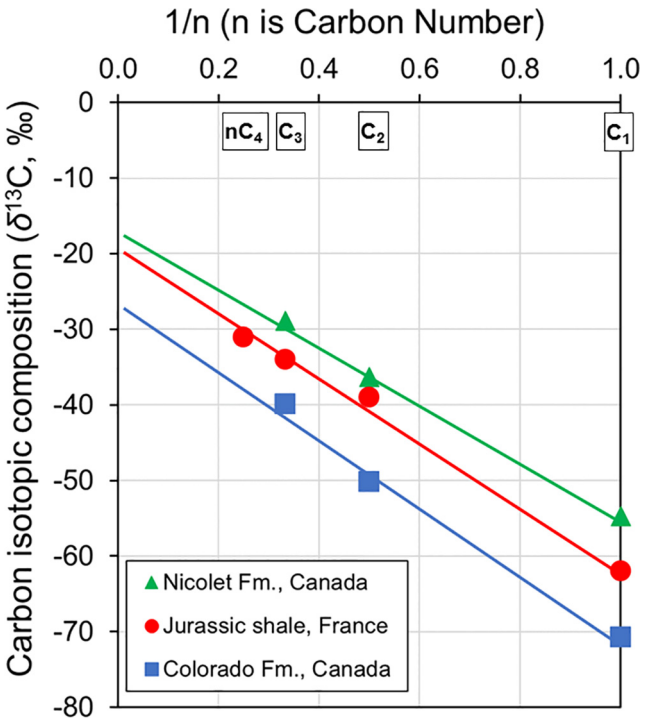

Fig. 10. Carbon isotopic composition ($\delta^{13}C$) of C_1–C_4 gases from shale formations in Canada (Rowe and Muehlenbachs, 1999; Lavoie et al., 2016) and France (Prinzhofer et al., 2009) displayed on the natural gas plot (Chung et al., 1988).

to the original C_1, depending on the isotopic compositions of original C_1 and CO_2 in the reservoir and the extent of secondary methanogenesis. For example, secondary microbial gas in the New Albany shale has C_1 more ^{13}C-enriched than the original very early mature thermogenic C_1 in that formation (Strąpoć et al., 2010) while biodegraded gas in the Barrow Island oil field in the North Carnarvon basin has C_1 more depleted in ^{13}C than C_1 in the feedstock non-biodegraded oil-associated thermogenic gas (Boreham and Edwards, 2008). Biodegradation and associated generation of secondary microbial gas result in complex changes in δ^2H-C_1 (Fig. 7D), which depend on the original C_1 present in the gas, the relative importance of various methanogenic pathways (CO_2-reduction, acetoclastic methanogenesis, methyloclastic methanogenesis etc., Dolfing et al., 2008; Feisthauer et al., 2010; Meslé et al., 2013) and the extent of the biodegradation in semi-open natural systems. As CO_2 derived from biodegraded oil is progressively converted into secondary microbial methane, the residual CO_2 becomes more enriched in ^{13}C (Fig. 9D; Milkov, 2018).

TSR leads to enrichment of gas in C_1, enrichment of C_1 in ^{13}C and ^{2}H, and enrichment of CO_2 in ^{13}C. However, as all these trends are consistent with increasing maturity of gases, additional gas components (especially H_2S) should be considered in the evaluation of TSR (e.g., Worden and Smalley, 2004). Molecular fractionation during migration may increase the $C_1/(C_2 + C_3)$ ratio but does not affect isotopic composition of C_1 (e.g., Etiope et al., 2009). Oxidation leads to depletion of gas in C_1 while the remaining C_1 becomes enriched in ^{13}C and ^{2}H reaching values of $\delta^{13}C$ and δ^2H as high as +45‰ and +301‰, respectively (Daskalopoulou et al., 2018).

The commonly occurring mixing of thermogenic and primary microbial gases was recognized on the original gas diagrams of Bernard et al. (1977) and Whiticar et al. (1986). Our study confirms that many natural gases are mixtures of gases with different sources and origin. Most gas samples located in the middle of the $\delta^{13}C$-C_1 versus $C_1/(C_2 + C_3)$ diagram ($C_1/(C_2 + C_3)$ from 50 to 1000 and $\delta^{13}C$-C_1 from −60‰ to −40‰, Fig. 7C, D) likely have complex origin and mixing history. For example, originally oil-associated thermogenic gas in many seeps and mud volcanoes, especially less active ones, display $C_1/(C_2 + C_3)$ ratio that resembles that of microbial gas, i.e. >500. This relative enrichment in C_1 is generally due to a combination of molecular fractionation during migration (e.g., Etiope et al., 2009) as well as preferential removal of C_{2+} gases and addition of secondary microbial C_1 during biodegradation (e.g., Milkov, 2011). Many samples of C_1 with $\delta^{13}C$ between −60‰ and −40‰ and δ^2H between −250‰ and −150‰ also have mixed origin (Fig. 8C, D).

As newly defined gas genetic fields often overlap on the individual gas diagrams (Fig. 7C, Fig. 8C, Fig. 9C; Supplementary material) and gas alteration and mixing further complicate interpretation (Fig. 7D, Fig. 8D, Fig. 9D), none of these diagrams should be used in isolation. Instead, gases should be plotted on all three diagrams discussed in this study and integrated interpretation should be performed. Furthermore, genetic diagrams utilizing gas compounds and isotopes not discussed in this study (N_2, H_2S, noble gases, δ^2H of C_{2-5} etc.) should be considered. Importantly, gas geochemical data should always be integrated with geological data describing gas habitat. Such holistic interpretation will result in more robust interpretation of gas origin and processes that affected gas composition.

6. Conclusions

In the present study, we revised three genetic diagrams ($\delta^{13}C$-C_1 versus $C_1/(C_2 + C_3)$, $\delta^{13}C$-C_1 versus δ^2H-C_1, and $\delta^{13}C$-C_1 versus $\delta^{13}C$-CO_2) most commonly used to interpret the origin of hydrocarbon-containing gases and processes that affect these gases after generation. The revised empirical diagrams are the most comprehensive among those published so far as they are based on a large (20,621 samples) dataset that includes gases from a wide variety of geographical and geological settings. Compared to previous versions, the new diagrams highlight the following:

1. The $\delta^{13}C$ range of thermogenic C_1 is from −75‰ to −15‰, as it includes early mature and very late mature thermogenic gases recently recognised in various petroleum systems. The thermogenic gases extremely enriched in ^{12}C or ^{13}C, however, are often mixed with microbial gases (in sediments at diagenetic to early catagenetic stages) and abiotic gases (e.g., in geothermal-volcanic plumbing systems involving organic-rich rocks).
2. Secondary microbial gases, produced during petroleum biodegradation and dominated by C_1, can be recognised from extreme enrichment of associated CO_2 in ^{13}C ($\delta^{13}C$ > +2‰ and, from currently available data, up to +36‰). Such significant ^{13}C-enrichment of CO_2 is rarely observed in gases of other origins.
3. Abiotic C_1 has a specific combination of $\delta^{13}C$ and δ^2H values and is not necessarily ^{13}C-eniched as was often considered in the past. The values of $\delta^{13}C$-C_1 can be as negative as −50‰, although a minor component of biotic (microbial or thermogenic) gas is often mixed with abiotic C_1.

The revised genetic diagrams provide a technically robust tool for gas interpretation, which can be used in various gas-related research projects, including both fundamental research (studies of subsurface fluid processes, carbon cycle and greenhouse gas emissions, origin of life etc.) and applied research (petroleum exploration/development, environmental geoscience etc.). For more reliable interpretations, inferences from gas genetic diagrams should be integrated with geological evidences and with geochemical interpretation based on other gas compounds such as non-hydrocarbon and noble gases.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.orggeochem.2018.09.002.

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